

# Data

Day 2 — The first-class citizen of every health AI problem

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# Where we are on the spine

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**Handoff** — two or three participants recap their Day 1 problem statement.

- **Yesterday:** an interesting topic → a well-posed problem.
- **Today:** a problem → the data that *actually exists* to attack it.

🔗 Day 1: problem before method

# The wrong frame: "we just need more data"

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- More data is **not** the goal — the **right** data is.
- Problems and data are **not independent**: the data you can obtain reshapes the problem you can pose.
- A data-blind problem statement is a wish, not a plan.

## The central claim

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Data is **not** a neutral input.

**It is generated by a process, for a purpose — and that shapes what you can credibly ask.**

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THE ORIGIN OF THE DATA

# The data-generating process

# Byproduct of care vs. designed for research

## Byproduct of care

- Generated to treat and to bill.
- Abundant, messy, biased by workflow.
- Most clinical data lives here.

## Designed for research

- Generated to answer a question.
- Rarer, cleaner *for its purpose* — still biased.
- Benchmarks, registries, trials.

**FIG F2.1 Two origins, two shapes.** A side-by-side contrast graphic: the same patient passing through "care delivery" (left) vs. a "research study" (right), with the different data each drops out. *See build guide F2.1.*

# The data is sampled from a *policy*

So provenance is a **selection mechanism**, not a list of quirks:

- Missingness is **MNAR by construction** — a value is absent because of a *decision* (not ordered  $\Rightarrow$  not worried). MCAR/MAR/MNAR labels are the floor; a **selection diagram** is the right object (Bareinboim & Pearl).
- "Controls" are defined by who the clinical policy *didn't* act on — selection, not sampling.
- Conditioning on a measurement (e.g. "lab was drawn") can open a **collider** and manufacture signal.

The data encodes the *policy* that produced it — and policies are not exchangeable across sites or time.

# Each modality is a measurement *decision*

| MODALITY           | THE MEASUREMENT IS ITSELF A DECISION... | ...WHICH BREAKS                           |
|--------------------|-----------------------------------------|-------------------------------------------|
| Labs               | ordered when a clinician is worried     | condition on "drawn" → collider bias      |
| Vitals             | sampled faster when acuity is high      | frequency is a severity proxy, not noise  |
| Notes              | dictated selectively, templated         | presence ≠ incidence; documentation drift |
| Dx / billing codes | coded for reimbursement                 | prevalence tracks coding incentives       |
| Imaging            | protocol- and referral-gated            | spectrum bias; site-specific acquisition  |



## Worked example — when the label is a billing artifact

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A diagnosis code looks like ground truth. It isn't.

- A code may appear because it **justifies reimbursement**, not because it's the primary clinical concern.
- It may be **absent** for a real condition that wasn't billed.
- Prevalence in the data tracks **coding incentives** as much as biology.

**Train on that label, and you partly learn the hospital's billing rules.**

🔄 Day 1: label noise

→ Day 3: estimands

# Case study — how a clinical dataset comes to exist

## MIMIC as a worked example:

- De-identified critical-care data from real patients (and **eICU-CRD** across many ICUs).
- **Built** through extraction, structuring, and a substantial de-identification effort.
- A "ready-to-use" dataset represents **years of invisible work** — and every choice in that work shapes the data.
- The **MIMIC Code Repository** exists because even reproducing a published result needs the shared, versioned cohort logic.

### FIG F2.2 From bedside to dataset. A

pipeline: bedside monitors / EHR →  
extraction → structuring → de-identification  
→ released dataset, with "shaping decisions"  
called out at each step. *See build guide F2.2.*

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GETTING THE DATA

# De-identification, governance, and access



# De-identification and governance

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- De-identification follows **standards** (Safe Harbor's 18 identifiers, expert determination) — not just deleting names.
- But de-id is **not a guarantee**. A model trained on the data can leak it: **membership inference** and memorization make re-identification a live risk — acute for clinical LLMs.
- The frontier tradeoff: **differential privacy** costs utility; **synthetic data** is *not* a privacy free lunch (it can still leak, and it shifts the distribution).

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Governance is a feature, not friction — the price of working with real patient data.

# Access mechanics — the practical path

- 1 **PhysioNet credentialing** — human-subjects training, an approved account.
- 2 **Data use agreement** — sign and comply with the DUA.
- 3 **IRB review** — know when your project needs it.

**FIG F2.3** **The access pipeline.** A simple numbered flow: training certificate → credentialed account → DUA signed → (IRB if needed) → data in hand. Mark a realistic elapsed-time bracket under each step. See build guide F2.3.

**Realistic timelines: weeks to months. Start early — before you need the data.**

# Standards solve syntax; representation is still open

- A common schema — **MEDS**, events as `(patient, time, code, value)` — makes data *portable*. Necessary, not sufficient.
- **How to represent** irregular, multimodal, multi-scale event streams is an **open research problem** — tokenization, time encoding, and code vocabularies are all contested.
- Standardization removes the busywork so the real modeling question (Day 4) is even askable.

**FIG F2.4 An event stream.** A patient timeline rendered as a sequence of (time, code, value) events — labs, meds, diagnoses on one axis. Foreshadows the modeling view on Day 4. *See build guide F2.4.*

→ Day 4: representation sets the method menu

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READING BETWEEN THE LINES

# What the data hides



# Provenance-driven bias

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- **Temporal bias** — practice and coding change; a model can learn the calendar.
- **Case-control pitfalls** — how cases and controls were selected can manufacture signal (Yuan et al.).
- **Selection effects** — who appears in the data, and who is **missing entirely**.
- **Coding & documentation artifacts** — a code's presence reflects workflow, not just physiology.



## Who is missing entirely

The most dangerous bias is in the patients who **never enter the data**.

- Under-served populations under-appear — or appear only at crisis.
- Missingness is **informative**: absence reflects access, not health.
- A model trained on who showed up encodes who **didn't**.

**FIG F2.5 The data vs. the population.** Two overlapping circles — "patients in the data" inside a larger "target population" — with the crescent gap labeled "never measured." *See build guide F2.5.*

→ Days 3 & 5: equity

→ data fusion (next)

## Frontier — "more data" can make it worse

- Combining sites is **data fusion**, not addition: each source carries its **own selection mechanism**.
- Pooling without modeling those mechanisms can **compound** bias — and *worsen* transport to a new site.
- "Transportable" is a claim about *which* mechanism is invariant — provable only under assumptions you must state (Bareinboim & Pearl, 2016).

The instinct "just get more data" is right only if you know what each source selected for.

🔗 Day 1: more sites ≠ more general

→ Day 5: transport

# Data answers the question it was generated for

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- Spending and utilization data reflect the **health system**, not just health (Einav et al.; Chandra et al.).
- There is almost always a **mismatch** between what the data *measures* and what you *want to know*.
- Name that gap explicitly — before it silently becomes your result.

## Worked example — spending is not health

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Predicting end-of-life spending (Einav et al., 2018):

- You can predict **spending** well — but spending tracks **treatment intensity and the system**, not just how sick someone is.
- "Who will be expensive"  $\neq$  "who is sick"  $\neq$  "who will benefit."
- A defensible-looking target can encode the system's choices, not the patient's needs.

 Day 1: cost vs. need

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TWO DATA WORLDS

# Institutional vs. patient-generated data



# The two data worlds

## Institutional / clinical

- Structured, episodic, billed.
- Captured *at* the point of care.
- Your lecture's focus.

## Patient-generated / sensor

- Continuous, consumer-grade, opt-in.
- Captured *away* from care.
- A different provenance, different biases.

**FIG F2.6 Two worlds.** A split panel: a hospital/EHR side vs. a wearable/phone side, each annotated with its sampling pattern and its characteristic bias (who is measured, when). See *build guide F2.6*.

## Designed-for-research data — a contrast

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- Purpose-built datasets (e.g. clinician-generated benchmarks, **MedAlign**) are **differently shaped** — and carry different biases, not none.
- They answer the question they were built for — and only that question.
- Even "clean" research data is a generating process you must interrogate.

# Interrogating your own dataset

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The four questions to ask of your data:

- 1 **Where did it come from** — what process generated it, for what purpose?
- 2 **Who is missing** — who never enters, who is under-measured?
- 3 **What does it silently encode** — coding, workflow, system incentives?
- 4 **Does it support your Day 1 problem** — or quietly change it?



# The gap check

Does the data that *exists* actually support the problem you posed yesterday?

- If **yes** — say how, specifically.
- If **no** — you must **revise the problem, the data plan, or both.**

A problem statement without a data spec is still a wish.

# Your data specification

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Interrogate the data your project actually depends on.

**Today's deliverable — Data specification sheet.**

The **generating process**, **what data exists**, your **access plan**, and the **anticipated provenance-driven bias**.

**Next:** 11:00 small group — interrogate your data, out loud · 1:30 guests (Sieberts & Banerjee, Sage Bionetworks) — patient-generated and research-cohort data.